

Deep mutational scanning and machine learning for the analysis of antimicrobial-peptide features driving membrane selectivity

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Many antimicrobial peptides directly disrupt bacterial membranes yet can also damage mammalian membranes. It is therefore central to their therapeutic use that rules governing the membrane selectivity of antimicrobial peptides be deciphered. However, this is difficult even for short peptides owing to the large combinatorial space of amino acid sequences. Here we describe a method for measuring the loss or maintenance of antimicrobial-peptide activity for thousands of peptide-sequence variants simultaneously, and its application to Protegrin-1, a potent yet toxic antimicrobial peptide, to determine the positional importance and flexibility of residues across its sequence while identifying variants with changes in membrane selectivity. More bacterially selective variants maintained a membrane-bound secondary structure while avoiding aromatic residues and cysteine pairs. A machine-learning model trained with our datasets accurately predicted membrane-specific activities for over 5.7 million Protegrin-1 variants, and identified one variant that showed substantially reduced toxicity and retention of activity in a mouse model of intraperitoneal infection. The high-throughput methodology may help elucidate sequence–structure–function relationships in antimicrobial peptides and inform the design of peptide-based synthetic drugs.

Antimicrobial peptides (AMPs), including host-defence peptides, are an evolutionarily conserved group of natural peptides that fight pathogens through direct antimicrobial activity and/or by modulating the host immune response during infection¹. AMPs generally kill bacteria directly through cell membrane disruption, but many show difficulties differentiating bacterial from mammalian cell membranes. This effect has been exploited in some cases where AMPs have been explored as anticancer agents. Some aspects of membrane selectivity have been described for alpha-helical AMPs². However, this information is limited and the sequence–activity relationships governing AMP membrane

selectivity remain largely unclear, which hinders their potential for therapeutic use.

The most thorough means of understanding AMP sequence–activity relationships is through mutational analysis. However, traditional methods for exploration of antimicrobial peptide residue importance and flexibility are slow and generally limited to small numbers of variants. One still popular method is to perform alanine scans by changing each position individually to alanine and measuring its effect on structure or activity. Such analogue methods limit the number and diversity of mutations that can be realistically examined. Newer strategies such

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as deep mutational scanning of protein sequences allow for large-scale assessment of how up to millions of mutations effect a measurable activity of interest but have not yet been as widely or successfully used with AMPs^{3,4}.

Recently, we developed a high-throughput method called surface localized antimicrobial display (SLAY), which has made it possible to assess the antimicrobial potential of hundreds of thousands of synthetic peptide sequences simultaneously⁵. Concurrent advancements in machine learning now allow models trained on known AMP datasets to predict the activities of untested sequences^{6,7}. We have previously trained machine learning algorithms with data gathered from biochemical validation of SLAY datasets to predict the antimicrobial activity for a library of ~200,000 peptide sequences with β -hairpin structure⁸. We now integrate deep mutational scanning with SLAY (dmSLAY) to allow the assessment of how thousands of amino acid changes to a known AMP sequence positively or negatively impact its antibacterial activity. We apply dmSLAY to Protegrin-1 (PG-1), a host-defence peptide originally identified in porcine leucocytes⁹, and determine positional residue importance and flexibility. We then use machine learning to predict both the antibacterial potency and toxicity of over 5.7 million PG-1 variants.

PG-1 is a member of a rare group of AMPs with β -hairpin structure cyclized via disulfide bonds. β -AMPs kill primarily via membrane lysis but show a wide range of antibacterial potency and mammalian toxicity which remains mostly unresolved¹⁰. PG-1 demonstrates potent and broad-spectrum bacterial membrane disruption^{9,11}, but it also shows high levels of haemolysis and cytotoxicity^{12,13}. Still, PG-1 has been successfully used as a scaffold by academics and biotechnology companies to design clinically relevant antibiotics such as Murepavadin^{14,15}. Although promising, phase III clinical trials with Murepavadin were ultimately terminated due to toxicity concerns¹⁶. Identifying the sequence features associated with the toxic side effects of similar β -AMPs could help fully realize the true clinical potential of this class. Applying dmSLAY to PG-1 uncovered rules governing its membrane selectivity and enabled us to train machine learning algorithms to digest millions of additional sequence variants and reveal important sequence–structure–function relationships influencing PG-1 bacterial and mammalian membrane selectivity. Our results establish dmSLAY as a method to gain more detailed insight into AMP sequence–activity relationships and reveal knowledge that could be used to improve rational optimization of β -AMPs to reduce their toxicity and advance their potential for clinical use.

Results

Deep mutational scanning of the Protegrin-1 sequence

We began our study of PG-1 by performing a traditional alanine scan in an effort to help elucidate residues that might be important for antimicrobial activity (Extended Data Fig. 1a). This scan, however, proved rather ineffective, since the only residue changes causing more than a twofold change in activity were R4A, Y7A and R19A, which were fourfold less potent as measured using the minimum inhibitory concentration (MIC).

To overcome the shortfalls of alanine-type scans and more efficiently elucidate PG-1 residue importance and flexibility, we developed deep mutational surface localized antimicrobial display (dmSLAY). To accomplish this, we first designed a genetically encoded deep mutational PG-1 variant library that was compatible with our antibacterial display approach. This library encoded 7 to 12 possible single-residue mutations at each residue position and 1 to 9 residue changes per unique variant (Extended Data Fig. 1b). In total, this library consists of 7,105 unique PG-1 sequences, 183 of which are single-residue variants. This is a massive increase in diversity over the 18 single mutants provided by traditional alanine scanning.

We next cloned the PG-1 variant library into an inducible surface display plasmid and transformed the library into *E. coli* W3110 for

high-throughput examination of antimicrobial potential using SLAY. SLAY exploits a bacterium's ability to both produce and simultaneously screen large peptide libraries for antibacterial potential. Each library is displayed on the outer membrane of bacteria by a tether fused to a fragment of OmpA. If a peptide is antibacterial, the bacterium expressing it is eliminated from the population only when expression is induced. Next-generation sequencing can quantify plasmids isolated from induced and uninduced cultures to generate a $\log_2$ (fold change) in reads predicting each peptide sequence's antimicrobial potential (Fig. 1a).

A SLAY screen run on our PG-1 variant library identified 1,940 probably inactive variants with a significant $\log_2$ (fold change) ($P < 0.05$) in reads greater than zero (Fig. 1b, blue). As expected, the native PG-1 sequence (PG-1.0) had a negative $\log_2$ (fold change) in reads. Using the PG-1.0 $\log_2$ (fold change) as a benchmark, another 1,203 variants were predicted to have more antimicrobial activity ($\log_2$ (fold change) < -1 , $P < 0.05$) (Fig. 1b, red). Variants with a significant $\log_2$ (fold change) score between 0 and -1 were deemed less active (Fig. 1b, grey). A principal component analysis of triplicate induced and uninduced samples showed clear grouping, distinct separation and high variance, suggesting that sample collection and analysis is consistent and reproducible (Extended Data Fig. 1c).

dmSLAY accurately predicts Protegrin-1 residue importance and flexibility

To investigate the accuracy of dmSLAY predictions, we synthesized 40 PG-1 variants spanning a range of $\log_2$ (fold change) scores and measured their MIC against *E. coli* W3110 in MH media (Supplementary Table 1). We also included data from 11 alanine scan variants already examined for which $\log_2$ (fold change) data were available. We used a receiver-operator characteristic curve to identify the optimal cut-offs for binning active and inactive variants using $\log_2$ (fold change) scores and MIC ($\text{MIC} \leq 64 \mu\text{g ml}^{-1}$; $\log_2$ (fold change) < 0) (Extended Data Fig. 1d). The $\log_2$ (fold change) score did not correlate directly with antibacterial potency, but a negative or positive score was 86% accurate at predicting whether a variant retained ($\text{MIC} \leq 64 \mu\text{g ml}^{-1}$) or lost ($\text{MIC} > 64$) antimicrobial activity (Fig. 1c). This included 16/17 accurate predictions for single-residue PG-1 variants examined in vitro (Fig. 1d, bold bordered cells) and indicated that dmSLAY was highly accurate at predicting whether a PG-1 variant retained or lost antibacterial activity. This analysis recategorized PG-1 variants with a significant $\log_2$ (fold change) from 0 to -1 as active (previously less active).

PG-1 has an antiparallel β -sheet structure consisting of two sheet regions with interstrand disulfide bonds, a loop region in between them and two tail regions near either terminus¹⁷ (Fig. 1d and Supplementary Fig. 1a). The tolerance of individual residue mutations based on significant $\log_2$ (fold change) from each dmSLAY single variant and the average $\log_2$ (fold change) for each individual PG-1 position (1–18) across all available residue changes is catalogued in a chart and colour coded (blue, grey or red) (Fig. 1d, far-left column). In total, these data encompass 47% of all possible single-residue variations. $\log_2$ (fold change) single variant dmSLAY data show that residues in the tail regions of PG-1 retain activity (red or grey), meaning that they are highly flexible, while many residue changes within the sheet regions, especially at position six, result in a loss of activity (blue), meaning that they may be generally more important for retention of antibacterial activity. It is important to note that many residue positions had amino acid changes predicted to either retain activity or lose activity, highlighting the importance of evaluating more than just one residue change at each position.

To evaluate mutations observed in PG-1 variants with multiple residue changes, a sequence logo plot was generated for the 50 variants with the lowest and highest dmSLAY $\log_2$ (fold change) scores (Supplementary Fig. 2b). Multiple residue variants predicted to lose activity often had mutations within the β -sheet regions, especially the

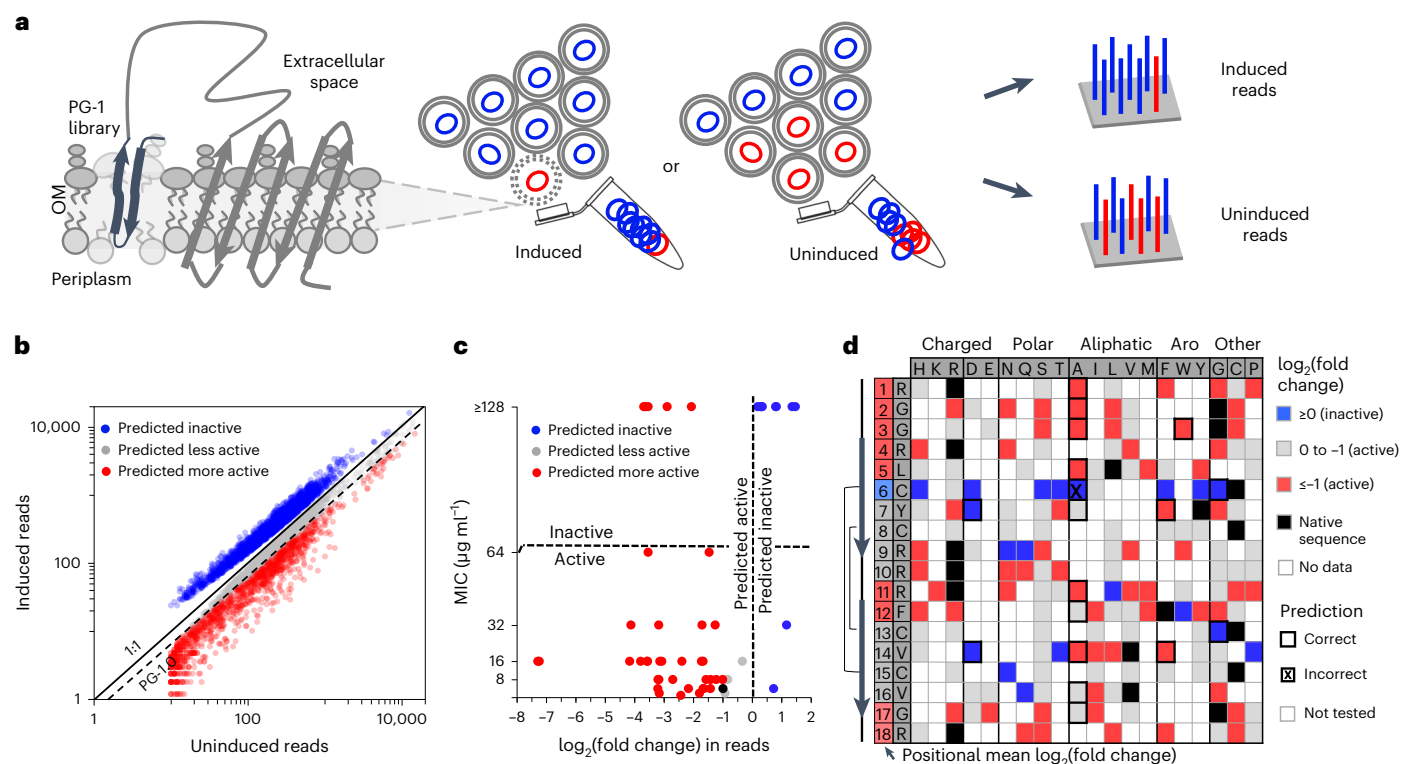


Fig. 1 | Protegrin-1 deep mutational SLAY predicts residue importance and flexibility. **a**, Surface localized antimicrobial display expresses an OmpA fragment tethering the PG-1 library to the outer membrane (OM). Induction of display results in cell death and lack of propagation for variants maintaining antimicrobial activity (red, active; blue, inactive). Change in reads between induced and uninduced cultures can predict antimicrobial potential. **b**, Scatterplot of variant mean induced versus uninduced reads ($n = 3$) with a significant $\log_2(\text{fold change})$ ($P < 0.5$) for the entire PG-1 library on log scales. The 1:1 and the native PG-1 sequence (PG-1.0) read ratios are shown. **c**, Scatterplot

of median MIC ($n = 3$) for a subset of 52 variants in the PG-1 library versus the $\log_2(\text{fold change})$ in reads. Optimized cut-offs for binning active and inactive variants are shown as dashed lines. **d**, Table charting dmSLAY predictions for all single PG-1 variants in the library. The native PG-1 sequence is indicated in column one, followed by amino acid change at each position going across the top categorized by side chain. PG-1 secondary structure including β -sheets (bold arrows) and disulfide bonds (brackets) are diagrammed at the left. Position is colour coded by $\log_2(\text{fold change})$. Bold boxed cells were evaluated in vitro. An X marks an incorrect dmSLAY prediction.

cysteine residues, consistent with observations from single variants. Variants retaining activity mostly contained changes in the tail regions, although some contained changes to sheet residues with similar side chain amino acids. These often also correlated with single variant data (that is, V14I or V14L).

Changes in secondary structure correlate with Protegrin-1 lytic activity

With our deep evaluation of residue importance in PG-1, we next looked to elucidate relationships between PG-1 sequence, structure and function. To do this, we used circular dichroism (CD) spectroscopy to examine changes in secondary structure for the 40 PG-1 variants selected above (Supplementary Table 1). Many AMPs, including PG-1, show an unstructured CD spectrum in buffer alone but fold into alpha-helical or β -sheet structure upon interaction with membranes or membrane mimics¹⁸. With this in mind, CD spectra were measured for PG-1.0 and all 40 variants with and without the presence of lipopolysaccharide (LPS), to mimic the bacterial outer membrane. As expected, native PG-1 and almost all its variants switched from unstructured (minimum at -200 nm) to β -sheet (minimum at -220 nm, maxima at -195) upon addition of LPS¹⁹ (Fig. 2a,b). To determine how secondary structure may impact antimicrobial activity, we colour coded spectra by MIC. Without LPS, we saw no discernible correlation between CD spectra and MIC (Fig. 2a); however, upon addition of LPS, we found that variants with a spectrum more representative of native PG-1 (PG-1.0) were generally more potent than those with larger changes in spectra (Fig. 2b). This suggests that the native secondary structure of PG-1

upon interaction with cell membranes is important for its antibacterial activity.

Because PG-1 kills bacteria by cell membrane lysis, we questioned whether this phenomenon was also true for lysis of mammalian membranes; therefore, we measured the haemolytic activity of each variant (Supplementary Table 1) and colour coded spectra by the percent haemolysis observed from treatment with $128 \mu\text{g ml}^{-1}$ peptide over 3 h (Fig. 2c,d). Again, we observed no correlation between percent haemolysis and CD spectra in the absence of LPS, but saw a similar correlation observed for MIC upon LPS addition, where haemolytic variants retained secondary structure close to PG-1.0 upon membrane interaction. One interesting difference was increased haemolytic spectrum with a maximum at 230 nm (Fig. 2b,d, arrows). This peak is found only in peptides containing tryptophan and is consistent with previous research showing that tryptophan residues generally increase mammalian cell toxicity^{20–22}.

Membrane selectivity is influenced by absence of large aromatics and loss of cysteine pairs

We noticed that some PG-1 variants appeared to have lost their haemolytic activity but remained strongly antibacterial, suggesting increased membrane selectivity (Fig. 2a,c, arrows). To better identify these selective variants, we generated a selectivity score for each variant tested by multiplying the MIC and the percent haemolysis. Variants with a lower selectivity score are more bacterially selective. We compared the selectivity score for all active PG-1 variants to PG-1.0 to identify those with improved membrane selectivity, resulting in five such variants

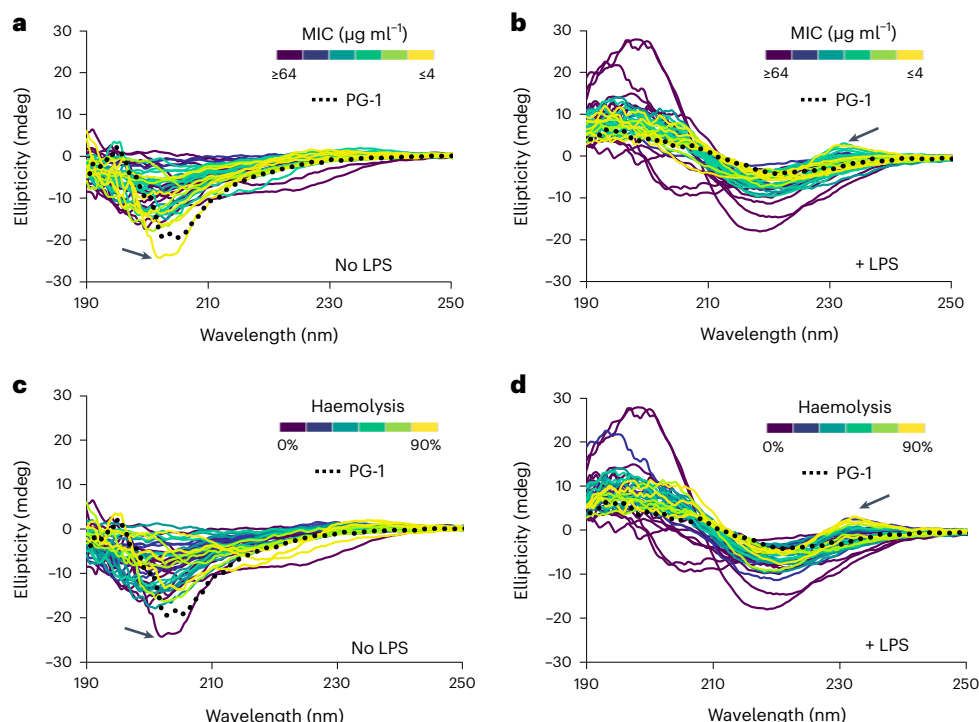


Fig. 2 | Changes in secondary structure correlate with Protegrin-1 lytic activity. **a–d**, Circular dichroism spectra of select PG-1 variants from the dmSLAY scan colour coded by minimum inhibitory concentration (**a,b**) or percent haemolysis (**c,d**). The native PG-1.0 sequence (PG-1.0) spectrum is shown as a

dotted line. Arrows indicate different trends in cell membrane lysis for PG-1.37 (**a,c**) or tryptophan containing variants with a maximum near 230 nm (**b,d**). All spectra are the average of technical triplicates.

(Fig. 3a). Selective variants were most heavily influenced by reductions in haemolytic activity (Extended Data Fig. 2a). We again colour coded CD spectra in the presence of LPS, this time by selectivity score, and found that spectra with a maximum at 230 nm were generally less selective except for PG-1.34 (Fig. 3b and Supplementary Table 1), but no other obvious correlations were observed. This is again consistent with peptides containing tryptophan residues having higher mammalian toxicity without impacting antibacterial activity. β -hairpin peptides are prone to aggregation which could also relate to their cellular toxicity. We tested whether toxicity correlated with the propensity of PG-1 variants to self-aggregate and found an exponential relationship ($R^2 = 0.63$) where highly haemolytic variants (>75%) were also highly aggregative and less haemolytic variants (<30%) were generally non-aggregative (Fig. 3c); however, there was a wider range of aggregation in variants with haemolytic activities between 30% and 75%, and the relationship did not fully explain differences observed in selectivity score.

Because changes in structure and self-aggregation did not explain differences observed in membrane selectivity, we broke down the sequence changes found in the four most selective variants to explore whether specific mutations influence selectivity (Fig. 3d). We found that mutation of phenylalanine at position 12 to either a serine or alanine accounted for the selectivity observed for PG-1.1, suggesting that phenylalanine itself may have a negative effect on selectivity. This supports our overall observation that large aromatics in PG-1 promote haemolysis and decreased selectivity. Unexpectedly, we identified cysteine mutants that retained antibacterial activity and improved their selectivity. Corresponding C6R and C15S mutations accounted for the increased selectivity of PG-1.20, whereas C15S alone matched PG-1.34's selectivity score; however, a corresponding mutation of C6G in PG-1.34 further reduced its selectivity score. These data suggest that removal of the disulfide bond between positions 6 and 15 to certain amino acids that do not strongly impact PG-1 structure may improve bacterial membrane selectivity. While most individual mutations of cysteine residues

are not well tolerated (Fig. 1d), insights gained through dmSLAY show that certain exceptions to this rule can improve selectively. Removal of the G3W mutation in PG-1.34 also caused a decrease in selectivity score, further supporting that tryptophan generally increases mammalian toxicity. PG-1.37 contains no changes to positions 6, 12 or 15 and its selectivity could only be recapitulated by the presence of all five mutations, suggesting that influences on selectivity for PG-1 variants with many mutations may be more complicated than simple removal of disulfide bonds and/or large aromatic residues; however, these variants can be identified using deep mutational scanning.

We noticed that all five more selective variants contained residue changes to serine and/or histidine (Supplementary Table 1). To see whether serine/histidine mutations have any general impact on selectivity, we synthesized additional PG-1 variants containing either a serine or histidine residue change with a significant negative log₂(fold change) from dmSLAY (Extended Data Fig. 2b). We used single variant predictions from dmSLAY to aid our selection (Fig. 1d). Five of the seven variants from this group had improved selectivity compared with native PG-1, but these improvements were more modest than those observed with PG-1.1, PG-1.20 and PG-1.37 (Extended Data Fig. 2c). Four of the five more selective variants contained mutations to the phenylalanine at position 12 or compatible mutations of two cysteines involved in a disulfide bond, suggesting again that removal of phenylalanine and/or compatible mutations to cysteines involved in a disulfide bond probably improves bacterial membrane selectivity (Extended Data Fig. 2b,c). PG-1.57 (V14A, R18H) and PG-1.37 were the only more selective PG-1 variants that did not contain these specific mutations.

Protegrin-1 variants demonstrate strong specificity for bacterial membranes

PG-1 kills bacteria via disruption of the outer and inner cell membrane. To ensure that our selective variants were killing via the same mechanism of action, we performed propidium iodide (PI) uptake assays.

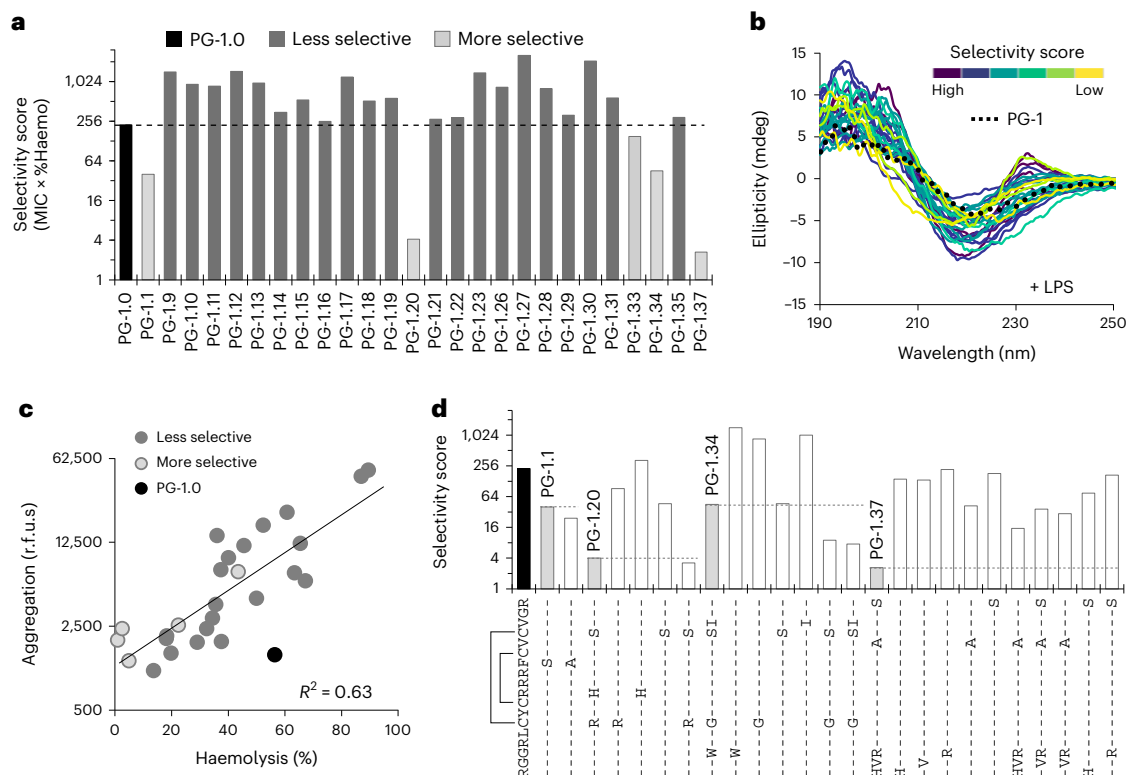


Fig. 3 | Membrane selectivity is influenced by aromatics and loss of cysteine pairs. **a**, Bar graph comparing selectivity score (median minimum inhibitory concentration × average percent haemolysis) for PG-1 variants confirmed to be antibacterial on a log scale. **b**, Circular dichroism spectra of same PG-1 variants colour coded by selectivity score. Spectra are the average of technical triplicates. **c**, Scatterplot of PG-1 variant aggregation in relative fluorescence units (r.f.u.s)

on a log scale versus linear percent haemolysis. Each data point is the mean of triplicate reactions; trendline represents a linear fit of log-transformed aggregation data versus haemolysis with $R^2 = 0.63$. **d**, Bar chart of selectivity score for a subset of mutations found in the four most selective PG-1 variants on a log scale. Brackets indicate where disulfide bonds are present in the native PG-1 sequence.

PI is a membrane impermeable dye that becomes fluorescent upon interaction with DNA; therefore, both the outer and inner membrane must be permeabilized for PI to access the cytoplasm and give a strong fluorescent signal. *E. coli* W3110 cells treated with serially diluted PG-1.0 for 30 min showed strong uptake of PI, as expected (Fig. 4a). Similarly, our selective variants also showed strong PI uptake, suggesting that they probably work via the same membrane lytic mechanism of action. Membrane lytic AMPs generally cause rapid cell death within 30 min of treatment. Kill curves performed with PG-1.0 and selective variants at concentrations fourfold their MIC show near complete *E. coli* W3110 cell death within 30 min, further supporting these variants killing via the same membrane lytic mechanism of action (Fig. 4b).

To examine in detail PG-1 variant bacterial membrane specificity, we simultaneously treated *E. coli* W3110 cells and human red blood cells with increasing concentrations of each variant. After 1 h of treatment, we measured both the percentage of bacterial killing and the percent haemolysis occurring at each concentration (Fig. 4c). Bacterial killing was first observed at $0.5 \mu\text{g ml}^{-1}$ for each variant. PG-1.20 and 1.37 showed no significant haemolytic activity, even at the highest concentration tested, while PG-1.1 and PG-1.0 began to noticeably lyse red blood cells at 128 and $8 \mu\text{g ml}^{-1}$, respectively, confirming previous observations when treated individually, suggesting that these variants are more toxic to red blood cells than PG-1.20 and PG-1.37. Comparing each variant's bacterial killing with and without the addition of red blood cells, we see no significant reduction in bacterial killing upon addition of 100-fold more red blood cells. This equates to greater than 1,000-fold more mammalian cell membrane surface area when considering differences in cell size (Extended Data Fig. 3). The inability of overwhelming amounts of mammalian cell membrane to

competitively inhibit antibacterial activity suggests that our variants contain greater than 1,000-fold more specificity for bacterial membrane compared with red blood cell membrane. Surprisingly, this also remained true for PG-1.0 despite the haemolysis observed at higher concentrations.

Machine learning identifies mutational profiles promoting membrane specificity

dmSLAY revealed several unexpected PG-1 variants with increased selectivity; however, these data are not sufficient for rational design of new PG-1 variants with desired qualities. Deep mutational scanning predicted the antibacterial potential of over 4,000 PG-1 sequence variants covering ~47% of possible single-residue changes, but it only assessed less than 0.1% of possible variants containing two or three residue changes. As seen with the selective PG-1 variants examined thus far, combinations of mutations can have a much greater effect on bacterial membrane selectivity than single mutations alone. To assess the full spectrum of possible one, two and three residue variants impacting function (>5.7 million), we trained three different machine learning models with our new biochemical dataset. These three algorithms focused on predicting the probability a PG-1 variant is highly potent ($\text{MIC} \leq 8 \mu\text{g ml}^{-1}$) (Extended Data Fig. 4a), predicting its haemolytic activity (Extended Data Fig. 4b) and predicting how selective it is towards bacterial membranes ($\log_{10}(\text{selectivity score})$) (Extended Data Fig. 4c). We trained each algorithm on 80% of our data and used the final 20% to validate the predictions made by the final models (Extended Data Fig. 4, top panels). A detailed description of the methods used to generate these models can be found in the Supplementary Methods. We then used these three models to identify the mutational profiles that have a high probability of being potent ($\text{MIC} \leq 8 \mu\text{g ml}^{-1}$), decrease

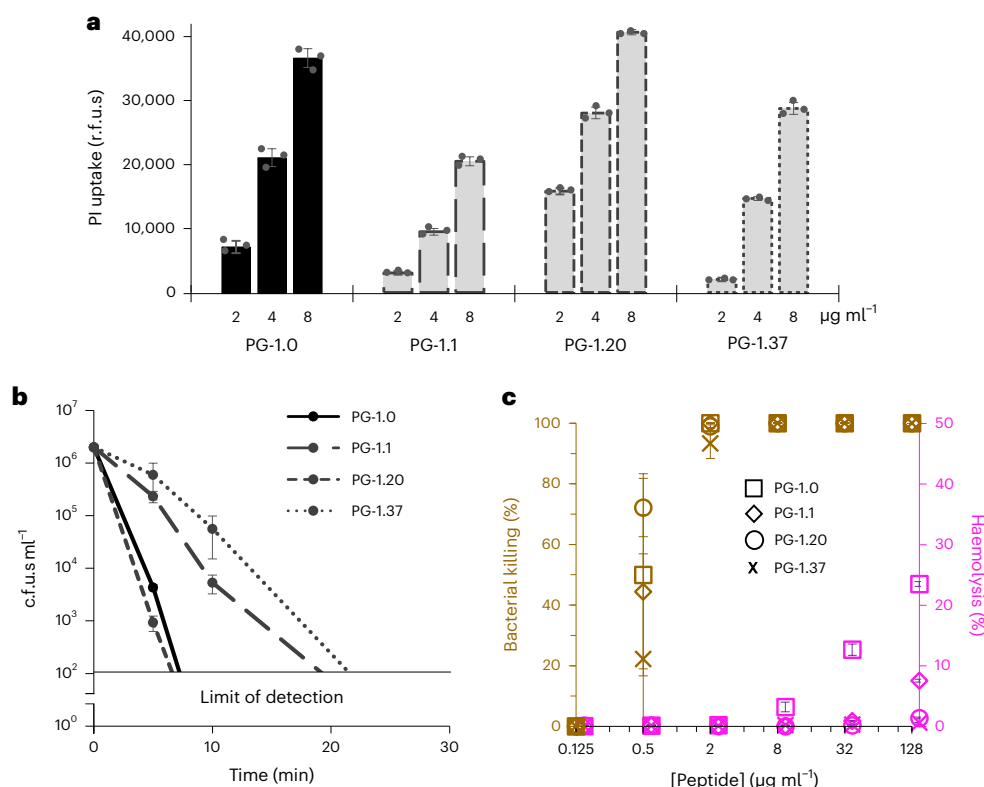


Fig. 4 | Protegrin-1 variants demonstrate strong specificity for bacterial membranes. **a**, Bar chart of PI uptake measured in r.f.u.s for *E. coli* cells treated with the indicated concentration of each PG-1 variant. Bars and dots represent the means $\pm$ 1 s.d. of biological triplicates. **b**, Kill curve showing c.f.u.s present over time for cultures treated with 8 $\mu\text{g ml}^{-1}$ peptide. **c**, Percentage of bacterial

killing and haemolysis observed in co-culture of *E. coli* and human red blood cells treated with increasing concentrations of PG-1 variants after 1 h of treatment. Data points in **b** and **c** are the means $\pm$ 1 s.d. of biological triplicates. Additional plots from individual and co-culture for each PG-1 variant are included in Supplementary Fig. 4.

mammalian toxicity (haemolysis $< 2\%$) or increase bacterial specificity ($\log_{10}(\text{selectivity score}) < 0.5$) individually. Heat maps of these mutational profiles generally show a preference for charged or polar residues in PG-1 variants retaining antibacterial activity. Changes are common in β -sheet residues for less haemolytic variants excluding tryptophan and cysteine. More membrane selective PG-1 variants show mutation of β -sheet residues to charged or polar amino acids while lacking addition of all aliphatic and aromatic residues, except alanine (Extended Data Fig. 4, bottom panels). Although these models provide detailed information regarding how PG-1 sequence impacts certain biochemical qualities, individually they do not select solely for highly potent variants specifically targeting only bacterial or mammalian membranes.

To find highly potent PG-1 variants that are non-haemolytic and bacterially specific, we combined all three models and identified 95,472 unique sequences passing all three cut-offs (Fig. 5a). To visualize the mutational profile of these bacterially selective PG-1 variants, we generated a heat map showing the number of times each mutation was observed within this bacterially selective group (Fig. 5b). Here, hydrophobic residues were changed within the β -sheet regions, especially at C6 and C15. These changes were often to polar or charged rather than other hydrophobic or aromatic residues with one notable exception being alanine, possibly due to its overrepresentation within the training dataset. Many of the most common residue changes observed did not exist in the training dataset at all, including C15T which was the most frequent change overall. For comparison, we also used our models to select for mammalian specific variants by adjusting the cut-offs used for each model to identify 51,876 mammalian specific variants (Fig. 5c). In their mutational heat map, we observed an opposite trend where hydrophobic or aromatic residues, especially tryptophan, replace mostly cationic residues in the loop and tail regions (Fig. 5d). Together,

this suggests that reducing hydrophobicity within the β -sheet regions of PG-1 generally improves bacterial membrane specificity while adding hydrophobicity, especially with tryptophan, in the tail and loop regions, whereas increasing overall charge improves mammalian membrane specificity. For a full list of variant sequences identified and their corresponding machine learning predictions for each model, see Supplementary Dataset 1.

Machine learning models identify bacterial specific PG-1 variants with reduced toxicity

To evaluate the performance of our machine learning models on an independent dataset, we selected 32 variants from the 95,472 predicted to be bacterially selective. To diversify the sequences found within this group, we randomly hand chose eight variants from four bins grouped by $\log_{10}(\text{selectivity score})$ and then measured the MIC and percent haemolysis of all 32 variants in vitro (Supplementary Table 2). We then used this information to calculate the selectivity score of each and compared attributes to see whether they were more bacterially selective than PG-1.0 and whether they met the cut-offs used in the machine learning predictions (Fig. 6a). From this group of 32 variants, 75% were active ($\text{MIC} \leq 64 \mu\text{g ml}^{-1}$) and all active variants had an improved selectivity score compared with PG-1.0. Their average $\log_2(\text{fold change})$ in selectivity score was an impressive -3.93 . In addition, 19% had an $\text{MIC} \leq 8 \mu\text{g ml}^{-1}$, 94% had under 2% haemolysis and 16% had a $\log_{10}(\text{selectivity score})$ of less than 0.5. For comparison, we contrasted these results to those obtained for the 31 dmSLAY active variants characterized during initial analysis of dmSLAY predictions (Supplementary Table 1, $\log_2(\text{fold change score}) < 0$). Here, 84% were active ($\text{MIC} \leq 64 \mu\text{g ml}^{-1}$), but only 16% were more selective than PG-1.0 and their average $\log_2(\text{fold change})$ in selectivity score was 1.68. Also,

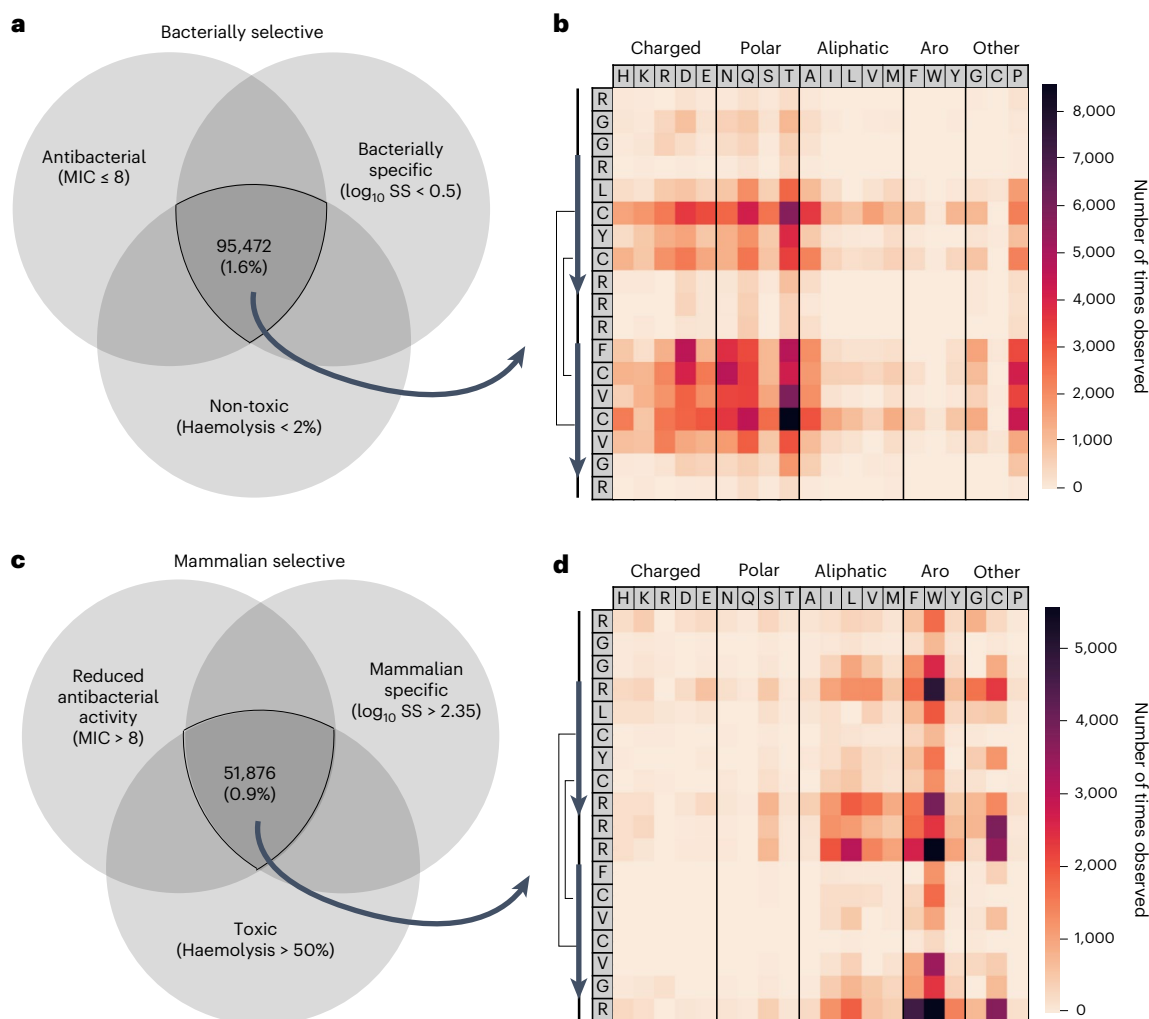


Fig. 5 | Machine learning identifies mutational profiles promoting membrane specificity. a, c. Venn diagram showing the three machine learning models trained on PG-1 variant data and the cut-off used to identify the 95,472 most bacterially specific (a) or 51,876 most mammalian specific (c) variants from over 5.7 million variants encompassing all possible one, two and three residue

mutations. **b, d.** Mutational heat maps charting the number of times each mutation is observed within the bacterially selective group (b) or mammalian selective group (d). The native PG-1 sequence is shown going down the far-left column and specific residue change across the top categorized by side chain.

29% had an MIC less than $8 \mu\text{g ml}^{-1}$, 6% had under 2% haemolysis and only 3% had a $\log_{10}$ (selectivity score) of less than 0.5. This means that our machine learning predictions vastly outperformed dmSLAY in identifying membrane selective and non-toxic PG-1 variants while being nearly as successful as dmSLAY in predicting antibacterial activity and potency.

When used as therapeutics, cationic AMPs commonly build up in the kidneys as they are removed from the blood, often resulting in tissue damage^{23,24}. To confirm that bacterially selective PG-1 variants were less toxic to kidney cells than native PG-1, we measured the toxicity of the best-performing dmSLAY and machine learning variants and compared them to PG-1.0 and Colistin, a clinically relevant AMP used to treat many Gram-negative antibiotic resistant infections (Fig. 6b). To do this, we treated a kidney tissue culture cell line (HEK293) with each AMP independently and measured HEK293 SYTOX Green uptake over time (Fig. 6b). SYTOX Green is a membrane impermeable dye used to assess mammalian cell membrane lysis. SYTOX Green only fluoresces upon DNA binding, meaning the cell membrane must be permeabilized. Both bacterially selective PG-1 variants had far less SYTOX Green uptake than PG-1.0, suggesting that they are less toxic to kidney cells. Interestingly, bsPG-1.2, selected through machine learning, also had less SYTOX Green uptake than Colistin at

each timepoint, suggesting that it may be less toxic to kidney cells if used therapeutically. Mobile Colistin resistance genes have recently evolved and spread as a means of generalized resistance to cationic AMPs by increasing the bacterial cell surface charge²⁵. Interestingly, PG-1.0 and our bacterially selective variants are unaffected by expression of mobile colistin resistance (*mcr-1*), suggesting that they are relatively unaffected by this common bacterial adaptation to AMP treatment (Supplementary Fig. 2a,b); however, PG-1 variant potency is still far weaker than clinically relevant antibiotics such as Colistin, which often include post-translational modifications or non-canonical amino acids.

Protegrin-1 naturally contains an amidated C terminus, so we next examined the effect of bsPG-1.2 amidation (bsPG-1.2 versus bsPG-1.2^A) on its activity against clinical strains and compared them to natural Protegrin-1 (PG-1^A) potency (Fig. 6c). We found that C-terminal amidation had a positive effect on bsPG-1.2 against *E. coli* ATCC 25922, but a negative effect on multidrug resistant *A. baumannii* AB5075 activity. Both bsPG-1.2 variants were more potent than Protegrin-1 at inhibiting *E. coli* ATCC 25922 growth, but only bsPG-1.2 was more active than Protegrin-1 on *A. baumannii* AB5075. Natural Protegrin-1 has been previously shown to reduce intravenous and intraperitoneal (IP) bacterial infections, but only relatively low doses ($\leq 5 \text{ mg kg}^{-1}$) have been

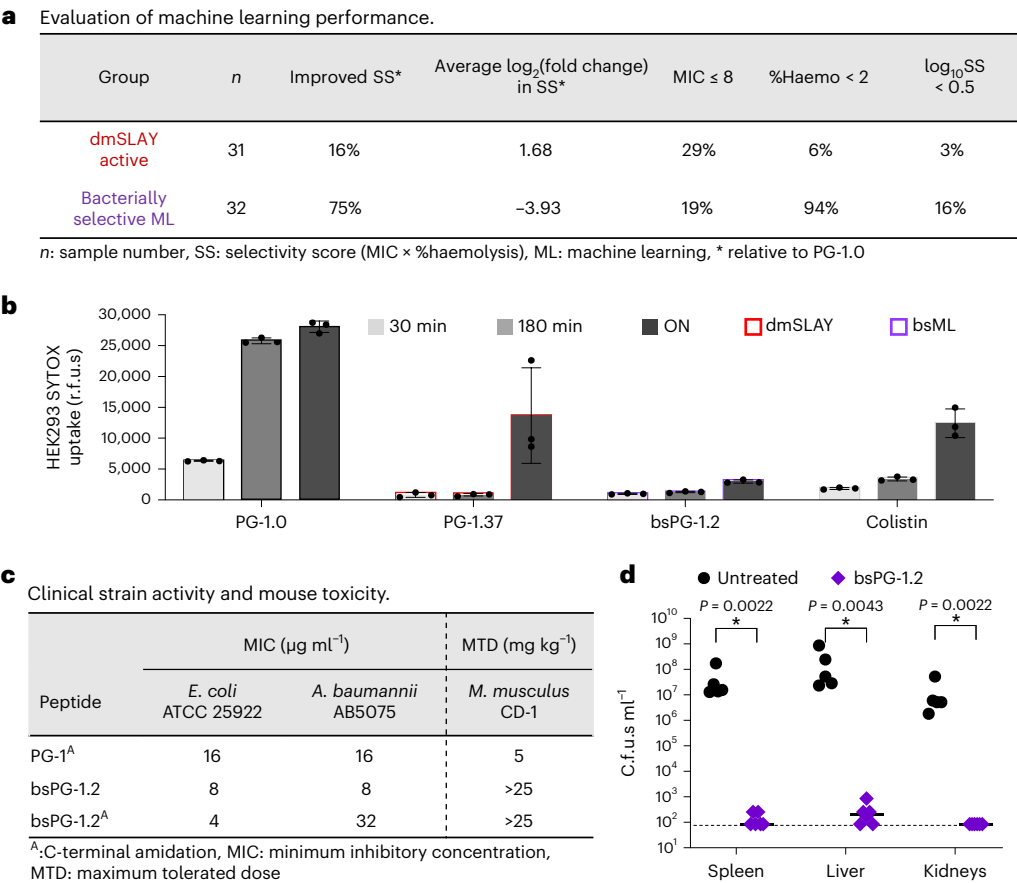


Fig. 6 | Analysis of machine learning performance. **a**, Table comparing characteristics of bacterially selective machine learning (bsML) and dmSLAY active PG-1 variants. **b**, Bar chart of SYTOX Green uptake (membrane lysis) of kidney cells (HEK293) treated with AMPs at three timepoints. ON, overnight. Bars and dots represent the means ± 1 s.d. of biological triplicates. **c**, Table showing the activity (MIC) and toxicity (MTD) of naturally occurring PG-1^A and bsPG-1.2

with and without C-terminal amidation. **d**, C.f.u.s present in the organs of CD-1 female mice at 6 h post intraperitoneal infection by injection of 10⁷ c.f.u.s of *A. baumannii* AB5075. Mice were immediately treated post infection by injection with either PBS (untreated, *n* = 5) or 25 mg kg⁻¹ bsPG-1.2 (*n* = 6). The limit of detection is indicated by a dashed line. Significance (*) was determined using multiple Mann–Whitney test.

used because of toxicity concerns^{11,26}. We determined the maximum tolerated dose (MTD) of Protegrin-1 and both bsPG-1.2 peptides by IP injection in CD-1 mice to further assess toxicity. The MTD of bsPG-1.2 and bsPG-1.2^A (>25 mg kg⁻¹) were more than fivefold greater than that of Protegrin-1 (5 mg kg⁻¹), suggesting that bsPG-1.2 is significantly less toxic to mice in vivo (Fig. 6c). bsPG-1.2 also demonstrated antibacterial activity in a mouse IP infection model (Fig. 6d). CD-1 mice were injected intraperitoneally with 10⁷ *A. baumannii* AB5075 colony-forming units (c.f.u.s) followed by immediate injection of either PBS (untreated) or a 25 mg kg⁻¹ dose of bsPG-1.2. Mice were left for 6 h, then euthanized and their organs dissected, homogenized and examined for the presence of bacteria by serial dilution plating for c.f.u.s. Organs from bsPG-1.2 treated mice had significantly less (*P* < 0.005) c.f.u.s present than organs from untreated mice. In fact, multiple mice treated with bsPG-1.2 had no detectable bacterial presence in any organ examined (c.f.u.s ml⁻¹ < 100).

Together, this work demonstrates that machine learning trained on biochemical data collected from dmSLAY-identified peptides can identify PG-1 variants with improved bacterial selectivity and reduced mammalian toxicity at a rate far surpassing low-throughput analogue methods and higher-throughput methods of peptide drug development, which are generally limited to less than 10,000 short peptides and are mostly inaccessible to research labs^{27–29}. Such techniques can be used to rapidly improve the therapeutic efficacy of known antimicrobial peptides for use as antibiotics.

Discussion

With the critical need for new antibiotics, increasingly more antimicrobial peptides have begun to enter the clinical pipeline¹. Mammalian cell toxicity continues to be a major barrier to their success, mostly due to our lack of knowledge regarding sequence–structure–function relationships influencing bacterial membrane specificity. This is largely driven by a lack of data, contradictory evidence and inconsistencies in sequence factors dictating AMP selectivity³⁰. Deep mutational SLAY analysis offers a high-throughput method of quickly determining positional residue importance and flexibility, while also predicting how antimicrobial activity is influenced by multiple simultaneous sequence changes (Fig. 1). The successful application of dmSLAY to PG-1 provides a compelling template for analysis of a wide range of AMP sequences and structures. This type of in-depth analysis across diverse unrelated sequences may help to reveal unifying AMP sequence–structure–activity relationships as well as highlight important differences unique to specific structural classes.

Our structural analysis of PG-1 variants revealed a strong correlation between secondary structure induced by membrane mimics and both bacterial and mammalian membrane lysis (Fig. 2). Previous studies have shown a similar correlation between changes in PG-1 secondary structure and cytotoxicity¹³ and correlations between antibacterial activity and haemolysis³¹; however, attempts to identify characteristics responsible for differentiation of these two activities have been generally unsuccessful¹². Cysteines have been previously identified as

important for retention of PG-1 antibacterial activity^{31–33}. Our dmSLAY results also support the importance of cysteine in maintaining antibacterial activity for single mutants; however, our work shows that specific corresponding mutations of cysteines involved in disulfide bonds which do not compromise native secondary structure can greatly reduce haemolysis and do so without impacting antibacterial potency (Fig. 3). These results demonstrate the importance of examining a more diverse set of sequence changes rather than just single point mutations when assessing AMP residue importance and flexibility.

While dmSLAY offers a greatly improved screening scale of AMPs, it can still only evaluate a small number of sequence variants compared with the total number of combinatorial possibilities. Here we amplified our exploration of the PG-1 sequence space by leveraging machine learning. This allowed the virtual evaluation of 5.7 million PG-1 variants for antimicrobial activity, toxicity and membrane selectivity. Machine learning methods are often trained on very large datasets. Remarkably, we developed highly accurate models of PG-1 activity using relatively few (<100) input sequences. Our modelling provided further insight into PG-1 sequence–activity relationships and suggests that reduction of hydrophobicity within the β -sheet regions while still maintaining secondary structure may be the primary driver of bacterial membrane selectivity (Fig. 4). This agrees with previous small analogue assessments of PG-1 sequence–activity relationships^{26,34}.

Previous studies using mostly alpha-helical peptides have shown that insertion of polar or charged residues into the hydrophobic face of AMPs can create imperfect amphipathicity and increased membrane selectivity³⁰. Highly hydrophobic or amphipathic peptides generally have a strong propensity for self-aggregation which has been shown to influence mammalian toxicity for some AMPs. Our PG-1 data also support a relationship between highly aggregative variants and increased toxicity. In addition, machine learning algorithms suggest that reducing overall charge and increasing hydrophobicity in the tail and loop regions, especially by mutation to tryptophan, generally increases mammalian membrane specificity. These changes would also be expected to increase the likelihood of aggregation.

One intriguing hypothesis is that PG-1 membrane selectivity is dictated by the localized concentration required to trigger self-aggregation at the cell membrane. Bacterial membranes are highly anionic, mammalian membranes less so. This would explain why PG-1.0 disrupts bacterial membranes at low concentration, but mammalian membranes require a significantly higher concentration. Increasing the critical concentration required to trigger self-aggregation by reducing peptide hydrophobicity and/or amphipathicity could prevent aggregation at the lower concentrations found near mammalian membranes but still trigger self-aggregation at increased local concentrations found near highly anionic bacterial membranes. This is something we will look to further explore in future studies.

Outlook

The strategy of combining dmSLAY with machine learning offers the ability to rapidly explore AMP sequence–structure–function relationships with far greater depth than ever before. This strategy can be applied to many AMP sequences to help elucidate rules governing AMP membrane selectivity, which can be used to inform the therapeutic optimization of more clinically relevant AMPs not amenable to dmSLAY, such as those requiring non-canonical amino acids and post-translational modification, to help us restock the currently bare antibiotic cupboard. These rules can also inform the synthetic design of peptide drugs treating a wider range of diseases requiring selective membrane permeabilization of different cell types to reach a variety of intracellular targets.

Methods

Minimum inhibitory concentration assays

Stock peptides were diluted to 256 $\mu\text{g ml}^{-1}$ in 350 μl of MH media, and 100 μl was aliquoted in the top row of a polypropylene 96-well plate

in biological triplicate. Peptides were then serially diluted twofold down columns of the plate in MH. *E. coli* W3110 were grown overnight in LB media at 37 °C. Cells from overnight cultures were diluted to a concentration of $\sim 1 \times 10^6$ cells per ml in MH, and 50 μl were added to each well of the 96-well plate containing diluted peptide. Plates were incubated at 37 °C for 18–24 h and examined by eye for growth. Final MIC is reported as the median of biological triplicate assays. Replicates never showed more than a twofold difference in MIC value.

Protegrin-1 library cloning

A library insert was first generated by PCR using forward primer oJR557 with reverse primer oJR560 and 2x(NR)tether gBlock as the template (Supplementary Table 3). The insert and the pMMBEH67_lpp_ompA vector were digested with KpnI and Sall and ligated overnight at 4 °C using T4 ligase. The ligated library was cleaned and transformed into *E. coli* W3110 competent cells via electroporation for further analysis via SLAY.

Deep mutational SLAY scan

E. coli W3110 frozen cells containing the BH plasmid library were diluted 1:1,000 and recovered in 10 ml of MH supplemented with 75 $\mu\text{g ml}^{-1}$ carbenicillin for 2 h. The culture was then back diluted to optical density (OD) of 0.05 and three biological triplicate 5 ml cultures were set up in MH supplemented with 75 $\mu\text{g ml}^{-1}$ carbenicillin. Triplicate reactions included: uninduced (0 μM IPTG) and induced (100 μM IPTG) cultures. All triplicate cultures were then grown for 4 h at 37 °C. Plasmids from each triplicate culture were then miniprep and Illumina sequencing primers were used to produce an amplicon via PCR (Supplementary Table 3). Amplicons were sent to Genewiz for next-generation sequencing by Illumina MiSeq technology with 30% added Phi-X DNA.

Deep mutational SLAY analysis

The reads obtained from next-generation sequencing underwent a meticulous series of filtering steps to isolate only the reads containing the desired amplicon regions. To achieve this, we initially utilized SeqKit³⁵, a powerful tool for sequence manipulation, to specifically select sequences that matched the desired amplicon regions. This ensured that only the relevant reads were retained for further analysis. Following the initial filtering step, we applied stringent criteria to exclude reads that did not meet the required standards. Specifically, reads with a quality score below Q30 were removed from the dataset. This quality threshold ensured that only reads with a high-accuracy sequence, indicative of reliable data, were retained for downstream analysis. To remove any remaining adapter sequences, we employed Flexbar³⁶, a versatile tool for read pre-processing, which we used to trim the reads, focusing solely on the variable region of interest—the actual peptide sequence that carries the biological information we sought.

Through the application of SeqKit and Flexbar, we isolated the desired amplicon regions and obtained high-quality reads with accurate peptide sequences. These processed reads formed the foundation for subsequent analyses, where we applied built-in programmes available in the Linux operating system. These programmes enabled us to efficiently sort and count the unique reads, providing us with information about the abundance and diversity of the peptide sequences within our dataset. The resulting counts of peptide reads were then utilized as input for DESEQ2 (ref. 37), a software package for differential gene expression analysis. By comparing the read counts between two libraries, DESEQ2 allowed us to identify peptides that exhibited significant differences in their read counts. This analysis provided us with insights into the variations in peptide abundance and expression levels, reported as $\log_2$ (fold change) scores, between the two libraries under investigation.

Peptide synthesis

All peptides used in this work were synthesized commercially by GenScript's custom peptide synthesis service and analysed by RP-HPLC

and mass spectrometry to confirm molecular weight. For high-purity peptides (>90%), final concentration was determined using the molecular weight, A_{205} and A_{205} extinction coefficient. Final concentrations of these peptides were also adjusted for purity. A full list of all the peptides used and their reported commercial purity can be found in Supplementary Dataset 1.

Haemolysis assay

Single-donor human red blood cells were washed in PBS and adjusted to a concentration of 1×10^9 cells per ml. Each peptide was added in biological triplicate to 200 μ l of cells at a concentration of 128 μ g ml⁻¹ in a 96-well polypropylene plate. PBS alone and 1% Triton X-100 were used for background normalization and 100% haemolysis, respectively. Plates were incubated for 3 h at 37 °C. Following incubation, samples were centrifuged at 800 g for 20 min and 100 μ l of supernatant was transferred to a flat-bottom 96-well plate. Percent haemolysis for each sample was determined by normalizing the absorbance at 540 nm for each sample to the average background and dividing by the average absorbance for 1% Triton X-100 (100% haemolysis). Error bars represent one standard deviation of triplicate samples.

Circular dichroism spectroscopy

Stock peptides were diluted in 10 mM phosphate (pH 7.4) to 200 μ g ml⁻¹ in a volume of 200 μ l. Samples were incubated for 1–2 h at room temperature and then analysed using a Jasco-815 CD spectrometer with a 0.1-cm-path-length quartz cuvette at the Targeted Therapeutic Drug Discovery and Development Program Core at the University of Texas, Austin. The CD spectra were collected using far-ultraviolet spectra (195–250 nm) with background corrected for 10 mM phosphate buffer alone. Reported spectra are an average of three separate spectra obtained from the same sample.

Aggregation assays

In a NUNC black-walled clear-bottom 96-well plate, PG-1 variants were diluted to 128 μ g ml⁻¹ in triplicate to a volume of 98 μ l in PBS. A volume of 2 μ l of prepared PROTEOSTAT detection reagent was then added to each well. The plate was then mixed for 10 s in a Biotek Synergy LX plate reader and allowed to incubate in the dark for 10 min, after which the fluorescence was read using the red filter cube (excitation 530/25, emission 590/35). Triplicate samples were normalized to PBS alone and reported as the mean.

Propidium iodide uptake

Single colonies from overnight growth on LB were inoculated into MH broth and grown to mid-log phase. Cells were then washed twice with PBS + 50 mM glucose and resuspended to an OD of 0.1 in PBS + 50 mM glucose. Propidium iodide was added in biological triplicate at a concentration of 10 μ g ml⁻¹, and 50 μ l of the propidium iodide cell mixture was quickly added to a pre-prepared 96-well plate (NUNC black-walled clear-bottom plate) that contained 50 μ l of various peptide concentrations. The plate was then allowed to incubate for 30 min in the dark at 37 °C and then read using the Biotek Synergy LX plate reader with the fluorescence red filter cube (excitation 530/25, emission 590/35). Triplicate samples were normalized to non-treated wells and presented as mean $\pm$ 1 s.d.

Kill curve

E. coli W3110 cells were back diluted from an overnight LB culture into MH to an OD of 0.001. Peptide stock solutions were diluted in MH to 8 μ g ml⁻¹. A volume of 100 μ l of diluted cells was combined with 100 μ l of diluted peptide stock in biological triplicate. Of each sample, 10 μ l were then serially diluted tenfold at the indicated timepoints and plated on LB agar. C.f.u.s were counted after overnight growth at 30 °C. Error bars represent one standard deviation of triplicate samples.

Protein representation and transfer learning

To ensure a robust representation of the peptides, we utilized transfer learning from the esm2 15 billion parameter model³⁸, a pre-trained protein Language Model (pLM), developed by Facebook. Features learned by the pLM can be transferred to any task (prediction) requiring a numerical representation (embeddings) by extracting these embeddings after the pre-training stage³⁹. This method enabled us to obtain numerical embeddings for each peptide sequence, resulting in a high-quality protein representation. The embedding extraction can be found on the esm2 GitHub page⁴⁰. Since the extracted embeddings contained more than 5,000 features, we needed to address the issue of overfitting. To mitigate this, we employed a feature selection technique to choose only those features that exhibited a high correlation with the target variable. We fine tuned the correlation threshold to maintain a minimal set of features, approximately equivalent to the number of samples in the training set. This approach allowed us to alleviate overfitting while retaining the most relevant and informative features.

Machine learning modelling

We have implemented a consensus machine learning model that determines whether a peptide sequence can be classified as a promising hit or a potential antimicrobial peptide. To achieve this classification, the peptide sequence must meet the cut-off thresholds of three distinct models trained to predict peptide activity, haemolysis and selectivity. To handle undefined values in the MIC model, such as >64 μ g ml⁻¹ MIC, they were converted to 100 μ g ml⁻¹ MIC. Subsequently, peptide values with MIC $\leq$ 8 μ g ml⁻¹ were classified as highly potent, while those above this threshold were classified as non-potent. These MIC-based classifications of peptides were utilized to train a classification model capable of predicting the probability that an antimicrobial peptide will be highly potent based on its amino acid sequence. In the haemolysis model, we utilized empirically obtained haemolysis percentages from laboratory experiments. These experimental values served as the target to train a regression model capable of predicting the percentage of haemolysis induced by each peptide. In addition, we introduced a new target variable called 'selectivity' by multiplying the MIC value by the haemolysis value and subsequently applying a logarithmic transformation ($\log_{10}$). This new selectivity score was then used to train a regression model capable of predicting the level of selectivity for each peptide. A lower selectivity score indicates an active peptide that is also non-haemolytic.

The dataset consisted of empirical data collected from 93 PG-1 variants (Supplementary Dataset 1). To train and evaluate our models, we divided this dataset into a training set, which encompassed 80% of the original data selected as a random sample, and a separate test set containing the remaining 20%. We employed various algorithms, including Random Forest, Support Vector Machine (SVM), Lasso and Decision Tree, among others, to train our models. To optimize their performance, we fine tuned the hyperparameters of these algorithms using a Grid search from the Scikit-learn library. For each target, we selected the best-performing model on the basis of its performance. Detailed information and implementation can be found on our GitHub page⁴¹ in the Jupyter Notebooks 04-06.

SYTOX Green uptake

Adherent HEK293 were grown to 80% confluence in Dulbecco's modified Eagle's medium supplemented with 10% fetal bovine serum and penicillin/streptomycin. Cells were then trypsinized, collected via centrifugation, washed twice and resuspended at 1.6×10^6 cells per ml in 1x Dulbecco's phosphate-buffered saline (DPBS). SYTOX Green was added to cells at 1:1,000 and then 50 μ l were added to a pre-prepared 96-well plate (NUNC black-walled clear-bottom plate) that contained 50 μ l of each peptide at 128 μ g ml⁻¹ in DPBS. The plate was then incubated at 37 °C in the dark and read using the Biotek Synergy LX plate reader with the fluorescence green filter cube (excitation 485/20, emission 528/20) at the indicated timepoints. Biological triplicate

samples were normalized to non-treated wells with no peptide added and presented as mean $\pm$ 1 s.d.

Mouse testing

All animal experiments were performed using approved Institutional Animal Care and Use Committee protocols. Female CD-1 mice (6–8-week-old) were used for all experiments. MTD studies were performed using ascending doses of peptide resuspended in PBS. The MTD was the highest dose that did not cause unacceptable side effects ($n = 2$). Intraperitoneal infection studies were performed as previously described⁴². Briefly, mice were rendered neutropenic by two doses of cyclophosphamide. A total of 11 mice were infected with 10^7 c.f.u.s of *A. baumannii* AB5075 in 100 μ l of PBS. Five mice immediately received a PBS injection (untreated) and six received a 25 mg kg⁻¹ dose of bsPG-1.2 dissolved in PBS. At 6 h after infection, mice were killed and the liver, spleen and kidneys were removed and homogenized. Serial dilutions (tenfold) of each homogenate were plated on LB agar with 75 μ g ml⁻¹ carbenicillin for bacterial enumeration. Prism was used to plot data and significance was determined with multiple Mann–Whitney test (* $P < 0.005$). Several samples were below the limit of detection (c.f.u.s ml⁻¹ < 100) and were set to 99 c.f.u.s ml⁻¹ for statistical analysis.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw sequencing data from dmSLAY are available from the SRA database under accession number [PRJNA1022479](https://www.ncbi.nlm.nih.gov/sra/PRJNA1022479). The raw and analysed datasets generated during the study are available for research purposes from the corresponding author on reasonable request. Source data for the figures are provided with this paper.

Code availability

Coding materials for sequence analysis and machine learning are available on GitHub at <https://github.com/facebookresearch/esm#main-models> ref. 40 and https://github.com/ziul-bio/DMS_ML_AMP ref. 41.

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Author contributions

J.R.R., L.C.V., B.W.D. and C.O.W. conceptualized this work. J.R.R. and L.C.V. were responsible for methodology, investigation and data visualization. B.W.D. and C.O.W. supervised the work and acquired funding. J.R.R. wrote the original draft and L.C.V., B.W.D. and C.O.W. helped with editing.

Competing interests

The authors declare no competing interests.

Additional information

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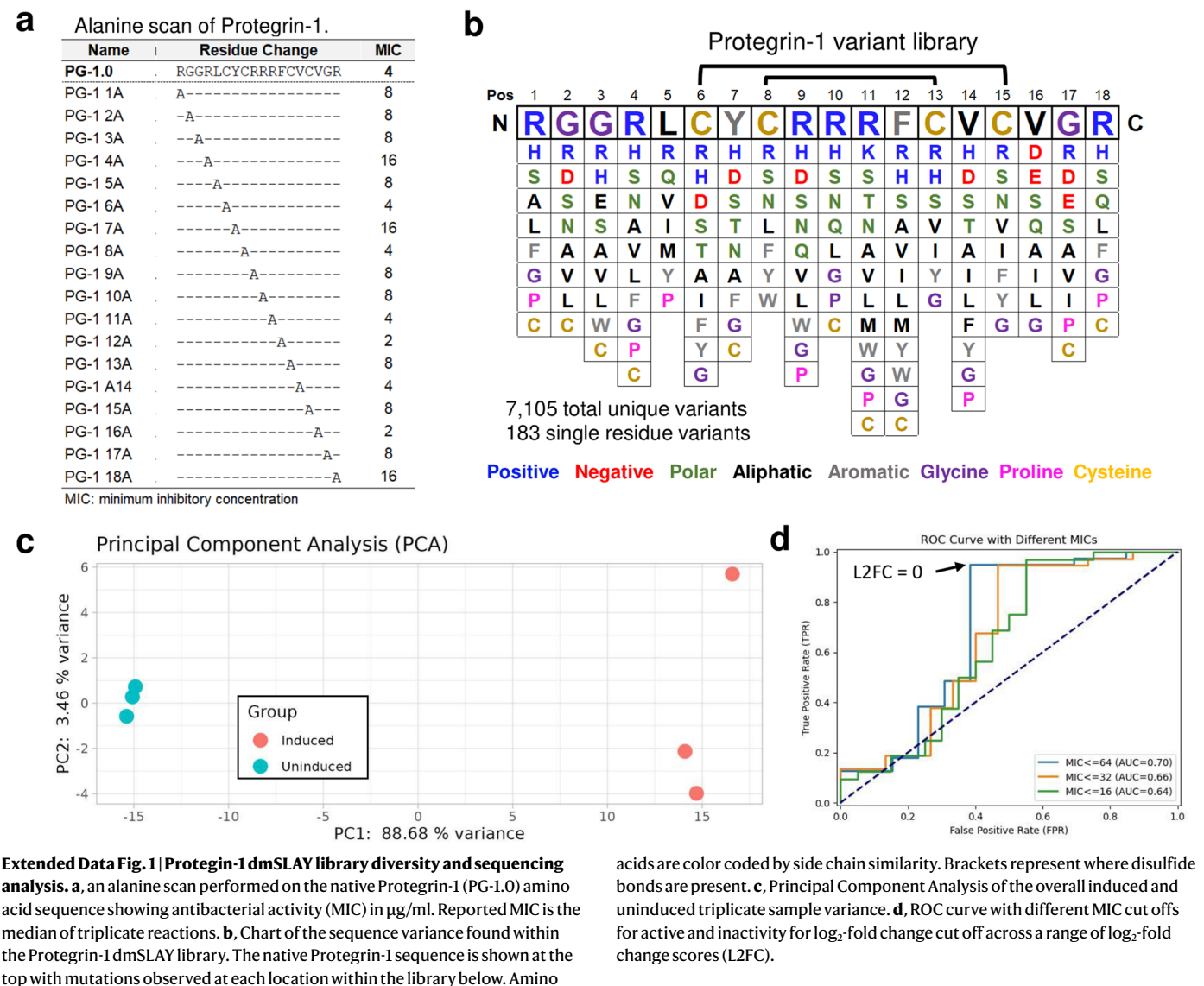
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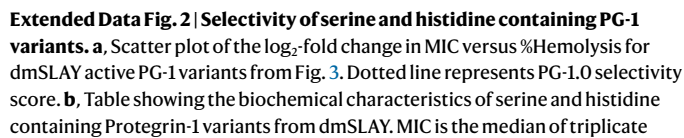
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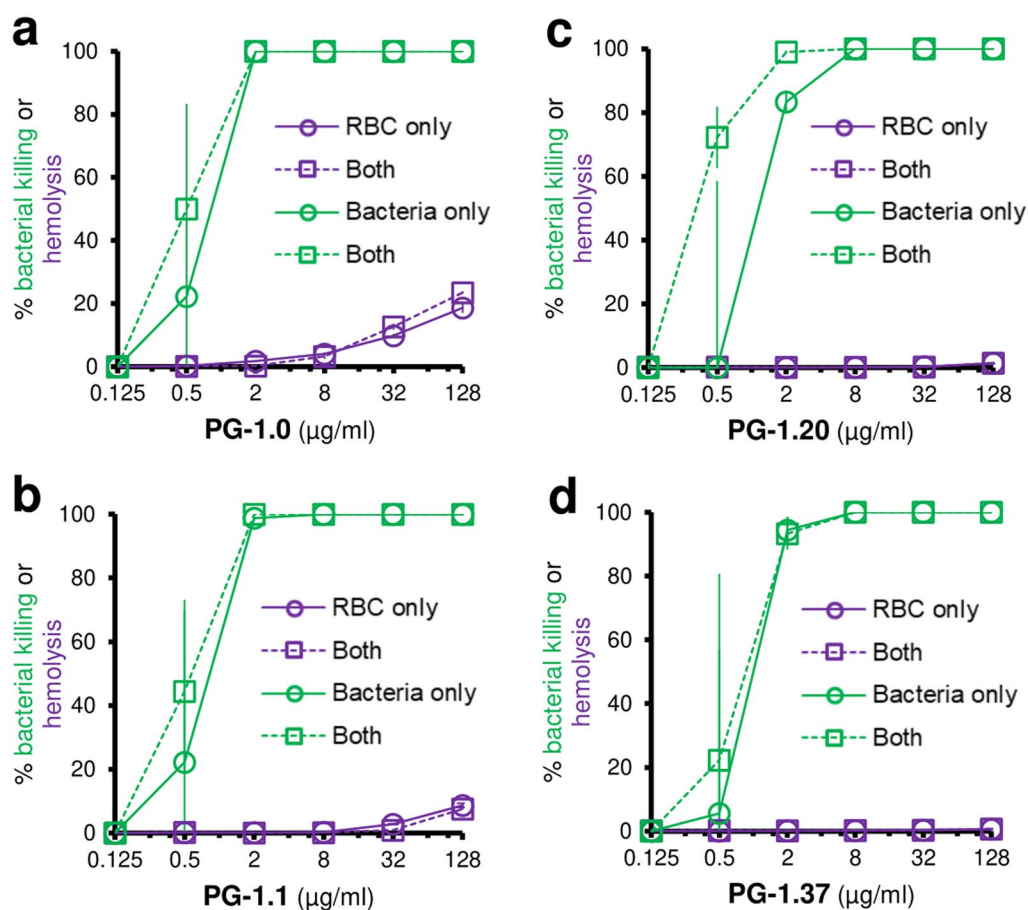


Extended Data Fig. 1 | Protegrin-1 dmSLAY library diversity and sequencing analysis. **a**, an alanine scan performed on the native Protegrin-1 (PG-1.0) amino acid sequence showing antibacterial activity (MIC) in $\mu\text{g/ml}$. Reported MIC is the median of triplicate reactions. **b**, Chart of the sequence variance found within the Protegrin-1 dmSLAY library. The native Protegrin-1 sequence is shown at the top with mutations observed at each location within the library below. Amino

acids are color coded by side chain similarity. Brackets represent where disulfide bonds are present. **c**, Principal Component Analysis of the overall induced and uninduced triplicate sample variance. **d**, ROC curve with different MIC cut offs for active and inactivity for $\log_2$ -fold change cut off across a range of $\log_2$ -fold change scores (L2FC).

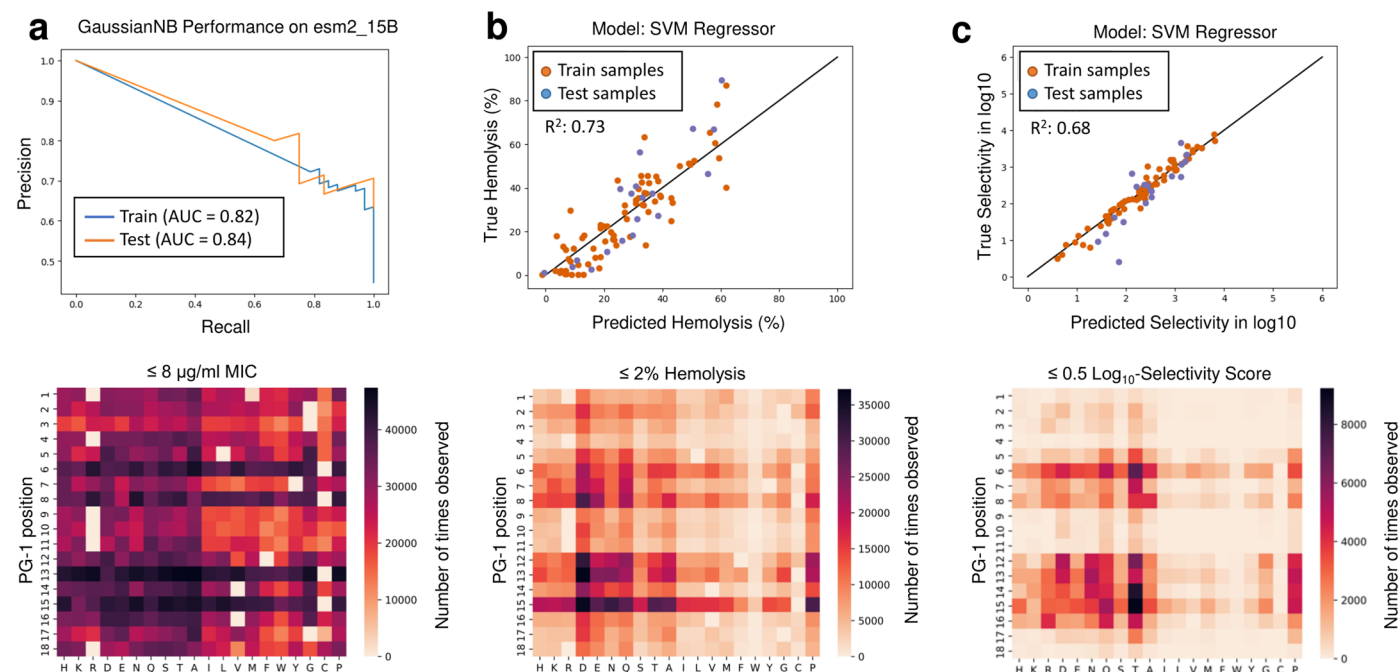


assays and %Hemolysis is the mean of triplicate assays. **c.** Bar chart showing the selectivity score of serine and histidine containing variants on a $\log_2$ scale. Residue changes are shown below. Brackets show where disulfide bonds are formed in the native structure.



Extended Data Fig. 3 | Comparing Protegrin-1 variant activity in mixed cultures. **a-d**, Graphs of PG-1 (top left), PG-1.1 (bottom left), PG-1.20 (top right), and PG1-37 (bottom right) percentage of bacterial killing (green) and % hemolysis

(purple) with 1×10^9 red blood cells (RBC), 1×10^6 E. coli W3110 cells (Bacteria) or both at various concentrations shown on a log₂ scale. Each data point is the mean of triplicate reactions and error bars are one standard deviation.



Extended Data Fig. 4 | Training of machine learning models and specific attribute mutational profiles. **a**, Precision and recall of for predicting PG-1 variants with an MIC $>$ or $< 8 \mu\text{g/ml}$. **b**, predicted versus true hemolysis for trained and test data **c**, or predicted versus true $\log_{10}$ -selectivity score. All models were

trained on 80% of data and validated with 20%. **Bottom panels:** Mutational profiles of variants from 5.7 million candidates with a predicted MIC $\leq 8 \mu\text{g/ml}$ (**a**) % hemolysis ≤ 2 (**b**) or $\log_{10}$ -Selectivity score ≤ 0.5 (**c**).

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Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s) Red Blood Cells (Innovative Research: IWB3ALS40ML), HEK293 (ATCC).

Authentication The HEK293 cells were not authenticated.

Mycoplasma contamination HEK293 cells were not tested for mycoplasma contamination.

Commonly misidentified lines (See [ICLAC](#) register) No commonly misidentified cell lines were used.

Animals and other research organisms

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Laboratory animals Mus Musculus CD-1 female mice, 6–8-weeks old.

Wild animals The study did not involve wild animals.

Reporting on sex Only female mice were tested.

Field-collected samples The study did not involve samples collected from the field.

Ethics oversight The animal experiments were performed with protocols approved by The University of Texas at Austin Institutional Animal Care and Use Committee.

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